

REAL-WORLD DATA PROJECT

Retail Store Sales Analysis Using Python

Submitted By

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Task Title:	Real-world Data Project (Retail Domain)

I. INTRODUCTION

Data Science plays an important role in analyzing real-world business data and generating meaningful insights for decision-making. Retail businesses use data analysis techniques to understand customer behavior, product performance, sales trends, and profit generation.

In this project, a retail sales dataset was analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. The project focused on identifying sales patterns, profit trends, and key business insights through statistical analysis and visualizations.

II. OBJECTIVE

- To analyze a real-world retail sales dataset
- To identify sales and profit trends
- To perform statistical and correlation analysis
- To visualize data using charts and heatmaps
- To generate meaningful business insights using data science techniques

III. TOOLS AND TECHNOLOGIES USED

Tools / Libraries	Purpose
Python	Programming Language
Pandas	Data Analysis and Manipulation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
VS Code	Code Editor and Execution

IV. DATASET DESCRIPTION

The dataset used in this project contains retail product sales information including product category, sales amount, profit, and quantity sold.

Sample Dataset

Product	Category	Sales	Profit	Quantity
Laptop	Electronics	50000	12000	5
Phone	Electronics	30000	8000	8
Shoes	Fashion	15000	4000	10
Watch	Accessories	12000	3000	7
Bag	Fashion	10000	2500	6
Tablet	Electronics	25000	6000	4
Headphones	Electronics	8000	2000	12
Tshirt	Fashion	7000	1500	15

V. METHODOLOGY

A. Importing Libraries

Required Python libraries such as Pandas, Matplotlib, and Seaborn were imported for data analysis and visualization.

B. Loading the Dataset

The retail dataset was loaded from a CSV file using the `read_csv()` function provided by Pandas.

C. Statistical Summary

Statistical analysis was performed using the `describe()` function to calculate mean, minimum value, maximum value, standard deviation, and count.

D. Sales and Profit Analysis

The total sales value was calculated, and the product with the highest profit was identified.

E. Correlation Analysis

Correlation analysis was performed to identify relationships between sales, profit, and quantity.

F. Data Visualization

Different visualizations were created to analyze the dataset effectively, including bar charts, scatter plots, and heatmaps. These visualizations helped in understanding business trends and product performance.

VI. OUTPUT

The retail sales dataset was successfully analyzed using Python libraries.

Key Findings

- Electronics products generated the highest sales
- Higher sales products produced higher profits
- Fashion products showed higher quantity sales with lower profit margins

Visualizations Generated

- Product vs Sales Bar Chart
- Sales vs Profit Scatter Plot
- Correlation Heatmap

Total Sales

157000

VII. RESULT

The analysis successfully identified important business trends and product performance patterns within the retail dataset. The project demonstrated how data analysis can help businesses make informed decisions regarding sales and profit optimization.

VIII. CONCLUSION

This project successfully demonstrated the application of Data Science techniques in a real-world retail business scenario. Statistical analysis and visualizations helped in understanding sales patterns, profit trends, and product performance. The project improved practical knowledge in data analysis, visualization, and

business insight generation.

IX. REFERENCES

- [Python Official Documentation](#)
- [Pandas Documentation](#)
- [Matplotlib Documentation](#)
- [Seaborn Documentation](#)